

Tutorial Week 7: Bayes Theorem

Part 1: Descriptive

1. Given HMM model represented by $M = (A, B, \pi)$. Describe the elements of a HMM.
2. Describe the forward recursion algorithm.

Part 2:

The below Multiple-Choice Questions (MCQs) focuses on “Hidden Markov Model”.

1. What are the possible values of the variable?
 - a) Variables
 - b) Literals
 - c) Discrete variable
 - d) Possible states of the world
2. Where does the Hidden Markov Model is used?
 - a) Speech recognition
 - b) Understanding of real world
 - c) Both speech recognition and understanding of real world
 - d) None of the mentioned

Part 3: Markov Model of Weather

Consider the weather condition is observable once a day Once a day as one of

S_1 : rainy S_2 : cloudy S_3 : sunny. The state transition probabilities A is:

$$A = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.1 & 0.8 \end{bmatrix}$$

1. Given the weather on day 1 ($t = 1$) is sunny(S_3), what is probability that the weather for the next 7 days will be sun-sun-rain-rain-sun-cloudy-sun?
2. Given that the model is state i , what is the probability that it stays in state i for exactly d days? What is the expected duration in state i (also conditioned on starting in state i)?

Part 1: Descriptive

1. - A is state transition probability matrix which shows the probability of moving from one hidden state to another hidden state.

e.g. $A_{ij} = P(q_{t+1} = s_j | q_t = s_i)$

- B is observation probability matrix which shows the probability of observing a certain output given a hidden state.

e.g. $B_j(u) = P(O_t = u_k | q_t = s_j)$

- π is initial state probability distribution which shows the probability of starting in each hidden state

e.g. $\pi_i = P(q_1 = s_i)$

2. - The forward recursion algorithm is used to calculate the probability of an observation sequence in a Hidden Markov Model.

- It works by computing the probability of reaching each state at each time step while considering all previous possible paths.

- The forward variable is

$$a_t(j) = P(O_1, O_2, \dots, O_t, q_t = s_j | M)$$

- The recursive variable is

$$a_{t+1}(j) = \left[\sum_i a_t(i) A_{ij} \right] B_j(O_{t+1})$$

- Steps: i) Initialisation

$$a_1(i) = \pi_i B_i(O_1)$$

- ii) Recursion

$$a_{t+1}(j) = \left[\sum_i a_t(i) A_{ij} \right] B_j(O_{t+1})$$

- iii) Termination

$$P(O|M) = \sum_i a_T(i)$$

Part 2: MCQ

1. d) Possible states of the world
2. c) Both speech recognition and understanding of real world.

Part 3: Markov Model of weather

States:

S_1 = rainy

S_2 = cloudy

S_3 = sunny

$$A = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.1 & 0.8 \end{bmatrix}$$

1. day 1: sunny

sun - sun - rain - rain - sun - cloudy - sun

$$\begin{aligned} P &= P(S_3 \rightarrow S_3) \times P(S_3 \rightarrow S_3) \times P(S_3 \rightarrow S_1) \times P(S_1 \rightarrow S_1) \times P(S_1 \rightarrow S_2) \times P(S_2 \rightarrow S_2) \times P(S_2 \rightarrow S_3) \\ &= 0.8 \times 0.8 \times 0.1 \times 0.4 \times 0.3 \times 0.1 \times 0.2 \\ &= 0.0001536 \\ &= 0.01536\% \end{aligned}$$

2.

Let $a_{ii} = P(S_i \rightarrow S_i)$

$$P(d) = (a_{ii})^{d-1} (1 - a_{ii})$$

$$E(d) = \frac{1}{1 - a_{ii}}$$

i) Rainy, S_1

$$a_{11} = 0.4$$

$$E(d) = \frac{1}{1 - 0.4}$$

$$= 1.67 \text{ days}$$

ii) Sunny, S_3

$$a_{33} = 0.8$$

$$E(d) = \frac{1}{1 - 0.8}$$

$$= 5 \text{ days}$$

ii) Cloudy, S_2

$$a_{22} = 0.6$$

$$E(d) = \frac{1}{1 - 0.6}$$

$$= 2.5 \text{ days}$$

$$\therefore E(S_1) = 1.67 \text{ days}$$

$$E(S_2) = 2.5 \text{ days}$$

$$E(S_3) = 5 \text{ days}$$